

Technology trends in India: From analogue to IP-and-video-analytic based surveillance in future

Current trend. IP based video surveillance is touted as the future of surveillance systems. It has replaced closed circuit analogue systems, which traditionally dominated the security market. IP-based systems are in demand with a trend towards remote surveillance and networkability.

By choosing IP based networks, building systems' operators get better response control, though the constraints would be availability of limited bandwidth, high cost of such systems and lack of volumes for this technology. Dua says, "Today's IP based surveillance cameras can handle motion detection and tampering detection. They have in-built intelligence that enables them to alert security personnel or to activate recording if an event takes place in the camera's surveillance area. Network video facilitates proactive monitoring."

"Thermal technology is also deemed one of the trends in the surveillance business," notes Dua. He adds, "The thermal network camera is a perfect complement to any professional IP surveillance system as it can be seamlessly combined with existing equipment, and it is possible to secure an area or a perimeter that lies in complete darkness."

Future trend. Apart from IP based, video analytic based surveillance is cited as one of the hottest technologies in the surveillance market.

Technology trends from around the world

Society of Motion Picture and Television Engineers (SMPTE) standard for SD, HD and 3G serial digital transmission. Considered to be one of the best trends, it enables security and surveillance systems to offer much better video quality. However, high-definition serial digital interface (HD-SDI) usage in the security and industrial markets has been limited by cable reach and the high cost of cabling infrastructure. IC solutions on the market have required significant cabling investment to realise the benefits of HD-quality video.

Fisheye image correction. Another trend is customers' increasing demands for acquisition of clear images. New fisheye image correction cameras process capabilities that enable the delivery of clear images, compensating for distortion and lighting. The technology includes a high-quality, high-resolution image signal processor (ISP) as well as an extremely flexible video output capability. Distorted hemispherical images generated by ultra-wide field of view of fisheye lens can be corrected and normalised into a rectangular output. Unlike software based solutions requiring a PC, the latest innovations have been designed specifically to be integrated into IP and analogue camera solutions.

Dirac or Vc-2 compression standards for long distance transmission. There are also emerging trends in compression standards, for example MPEG4 and H.264, which offer advantages and challenges. Image compression allows video to be transmitted over longer distances, but can result in a reduction in quality. Dirac or Vc-2 compression enables HD video to be transmitted over long distances or over lower-cost cables even in noisy environments.

Internet of Things (IoT) impacting the surveillance and security market. IoT is connecting devices inside the home and within local networks to add intelligence and share information broadly across devices. The invention of brand new use cases and applications where the ability to transmit video is a key feature has exploded. Inexpensive, application based nanny cams and security systems are becoming well accepted among consumers. Although, the most significant demands introduced by the Internet of Things are power/battery management and footprint.

— Contributed by Joe Hanson, product marketing engineer, Intersil

network videos into their enterprise IT-security plans.

Himadri Roy, assistant manager, Netscribes Inc., says, "The market for private security services has exhibited the fastest growth among the rest of the segments, followed by surveillance and monitoring, and electronic access control equipment," He further adds,

"It is anticipated that the government of India will invest close to ₹ 554 billion (US\$ 9 billion) toward private security services by the end of 2016."

Lower system costs and reduced manpower costs due to increased intelligence also provide a lucrative market. Dua believes, "Besides government programs that are designed

to strengthen borders and inhibit terrorism, this growth is also fuelled by the ability of electronics to reduce manpower and increase effectiveness of systems."

The overall market faces maximum demand from key industry verticals such as healthcare, real estate and BFSI, believes Roy. He says, "All these industries are poised for a steady growth in the next four to five years, as a result of which the demand curve for security equipment and services is expected to shape up positively as well." He adds, "So far the market has not seen much of sales from the online space, thereby creating significant opportunity areas for players, especially when the state of Internet connectivity and Internet user base is improving exponentially on a daily basis."

Growing demand for IP-based surveillance

The internet protocol (IP) based or digital network video surveillance technology has emerged as a fast-growing technology, attracting small and medium scale integrators, and resellers into this sector.

"Over the past few years, we have seen higher adoption rates for IP surveillance over legacy analogue deployments. The Indian customer today is well informed and wants to invest in the best available technology," says Prakash Prabhu, general manager, Pelco, Schneider Electric India. He adds, "The surveillance industry, per se, has gone through multiple technological changes in terms of adopting technology, namely, H.264 compression standard, megapixel (MP) to high definition (HD) resolution and server-to-onboard analytics."

The companies in India are likely to increase their security budgets by 35 to 40 per cent. Statistically speaking, Dua shares, "The Indian surveillance market, which constitutes about 35 per cent of the security industry, is seeing a shift from analogue to IP-based surveillance." He adds, "Currently, IP-based surveillance market in India is expected to show year-on-